$$\begin{cases}3x_1+x_2+2x_3\xrightarrow{18}\max\\ x_1+x_2+x_3\le 10\\ 2x_1+x_3\le 8\\ x_2+3x_3\le 15\\ x_1,x_2,x_3\ge 0\end{cases}\qquad \begin{cases}-3x_1-x_2-2x_3\xrightarrow{-18}\min\\ x_1+x_2+x_3+x_4=10\\ 2x_1+x_3+x_5=8\\ x_2+3x_3+x_6=15\\ x_1,x_2,x_3,x_4,x_5,x_6\ge 0\end{cases}$$

$$\begin{array}{c|cccccc|c} & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & b\\\hline x_4 & 1 & 1 & 1 & 1 & 0 & 0 & 10\\ x_5 & 2 & 0 & 1 & 0 & 1 & 0 & 8\\ x_6 & 0 & 1 & 3 & 0 & 0 & 1 & 15\\\hline F & -3 & -1 & -2 & 0 & 0 & 0 & 0\end{array}\quad \frac{b}{x_1}\quad \frac{10}{1}>\frac{8}{2}\quad :2\quad \cdot 3/2$$

$$\begin{array}{c|cccccc|c} & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & b\\\hline x_4 & 0 & 1 & \frac12 & 1 & -\frac12 & 0 & 6\\ x_1 & 1 & 0 & \frac12 & 0 & \frac12 & 0 & 4\\ x_6 & 0 & 1 & 3 & 0 & 0 & 1 & 15\\\hline F & 0 & -1 & -\frac12 & 0 & \frac32 & 0 & 12\end{array}$$

$$=\begin{array}{c|cccccc|c} & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & b\\\hline x_2 & 0 & 1 & \frac12 & 1 & -\frac12 & 0 & 6\\ x_1 & 1 & 0 & \frac12 & 0 & \frac12 & 0 & 4\\ x_6 & 0 & 0 & \frac52 & -1 & \frac12 & 1 & 9\\\hline F & 0 & 0 & 0 & 1 & 1 & 0 & 18\end{array}$$

$x_1=4;\ x_2=6;\ x_6=0$

$x_3=0;\ x_4=0;\ x_5=0$

$F=18$

$$\begin{cases}-3x_1-x_2-2x_3\to\min\\ x_1+x_2+x_3+x_4=10\\ 2x_1+x_3+x_5=8\\ x_2+3x_3+x_6=15\\ x_1,x_2,x_3,x_4,x_5,x_6\ge 0\end{cases}\qquad \begin{cases}x_7+x_8+x_9\to\min\\ x_1+x_2+x_3+x_4+x_7=10\\ 2x_1+x_3+x_5+x_8=8\\ x_2+3x_3+x_6+x_9=15\\ x_1\ldots x_9\ge 0\end{cases}$$

$$\begin{array}{c|cccccc|c} & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & b\\\hline x_7 & 1 & 1 & 1 & 1 & 0 & 0 & 10\\ x_8 & 2 & 0 & 1 & 0 & 1 & 0 & 8\\ x_9 & 0 & 1 & 3 & 0 & 0 & 1 & 15:3\\\hline -\Sigma & -3 & -2 & -5 & -1 & -1 & -1 & -33\end{array}\quad \cdot 5/3 \quad = \quad \begin{array}{c|cccccc|c} & x_1 & x_2 & x_9 & x_4 & x_5 & x_6 & b\\\hline x_7 & 1 & \frac23 & 0 & 1 & 0 & -\frac13 & 5\\ x_8 & 2 & -\frac13 & 0 & 0 & 1 & -\frac13 & 3:2\\ x_3 & 0 & \frac13 & 1 & 0 & 0 & \frac13 & 5\\\hline -\Sigma & -3 & -\frac13 & 0 & -1 & -1 & \frac23 & -8\end{array}\quad \cdot\frac32\quad =$$

$$=\begin{array}{c|cccccc|c} & x_8 & x_2 & x_9 & x_4 & x_5 & x_6 & b\\\hline x_7 & 0 & \frac56 & 0 & 1 & -\frac12 & -\frac16 & \frac72\\ x_1 & 1 & -\frac16 & 0 & 0 & \frac12 & -\frac16 & \frac32\\ x_3 & 0 & \frac13 & 1 & 0 & 0 & \frac13 & 5\\\hline -\Sigma & 0 & -\frac56 & 0 & -1 & \frac12 & \frac16 & -\frac72\end{array}\ =\ \begin{array}{c|cccccc|c} & x_8 & x_2 & x_9 & x_7 & x_5 & x_6 & b\\\hline x_4 & 0 & \frac56 & 0 & 1 & -\frac12 & -\frac16 & \frac72\\ x_1 & 1 & -\frac16 & 0 & 0 & \frac12 & -\frac16 & \frac32\\ x_3 & 0 & \frac13 & 1 & 0 & 0 & \frac13 & 5\\\hline -\Sigma & 0 & 0 & 0 & 0 & 0 & 0 & 0\end{array}\ =$$

$$=\begin{array}{c|ccc|c} & x_2 & x_5 & x_6 & b\\\hline x_4 & \frac56 & -\frac12 & -\frac16 & \frac72\\ x_1 & -\frac16 & \frac12 & -\frac16 & \frac32\\ x_3 & \frac13 & 0 & \frac13 & 5\\\hline -\Sigma & 0 & 0 & 0 & 0\end{array}$$

$x_4+\frac56x_2-\frac12x_5-\frac16x_6=\frac72;$

$x_1-\frac16x_2+\frac12x_5-\frac16x_6=\frac32;\ x_1=\frac32+\frac16x_2-\frac12x_5+\frac16x_6$

$x_3+\frac13x_2+\frac13x_6=5;\ x_3=5-\frac13x_2-\frac13x_6$

$-3x_1-x_2-2x_3\to\min;\ -3\left(\frac32+\frac16x_2-\frac12x_5+\frac16x_6\right)-x_2-2\left(5-\frac13x_2-\frac13x_6\right)=$

$=-\frac92-\frac12x_2+\frac32x_5-\frac12x_6-x_2-10+\frac23x_2+\frac23x_6=-\frac56x_2+\frac32x_5+\frac16x_6-\frac{29}{2}$

$$\begin{array}{c|ccc|c} & x_2 & x_5 & x_6 & b\\\hline x_4 & \frac56 & -\frac12 & -\frac16 & \frac72\cdot\frac65\\ x_1 & -\frac16 & \frac12 & -\frac16 & \frac32\\ x_3 & \frac13 & 0 & \frac13 & 5\\\hline F & -\frac56 & \frac32 & \frac16 & \frac{29}{2}\end{array}\quad :5\quad \cdot\frac25\quad =\quad \begin{array}{c|ccc|c} & x_4 & x_5 & x_6 & b\\\hline x_2 & 1 & -\frac{6}{10} & -\frac15 & \frac{42}{10}\\ x_1 & 0 & \frac{4}{10} & -\frac15 & \frac{22}{10}\\ x_3 & 0 & -\frac15 & \frac{4}{10} & \frac{36}{10}\\\hline F & 0 & 1 & 0 & 18\end{array}$$

$x_1=2{,}2;\ x_2=4{,}2;\ x_3=3{,}6$

$x_4=0;\ x_5=0;\ x_6=0$

$F=18$

$$\begin{cases} 4x_1+6x_2+5x_3+3x_4\to max\\ x_1+2x_2+x_3+x_4=12\\ 2x_1+x_2+3x_3+x_4=18\\ x_1,\dots,x_4\geq 0\end{cases}\quad \begin{cases} -4x_1-6x_2-5x_3-3x_4\to min\\ -//-\end{cases}\quad \begin{cases} x_5+x_6\to min\\ x_1+2x_2+x_3+x_4+x_5=12\\ 2x_1+x_2+3x_3+x_4+x_6=18\\ x_1\dots x_6\geq 0\end{cases}$$

$$\begin{array}{c} \\ x_5\\ x_6\\ -\Sigma \end{array}\begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & b \end{array}\\ \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 12\\ 2 & 1 & 3 & 1 & 18\\ -3 & -3 & -4 & -2 & -30 \end{array}\right)\end{array} \begin{array}{l} \\ \\ :3\ \ \cdot\frac{4}{3}\\ \\ \end{array} = \begin{array}{c} \\ x_5\\ x_3\\ -\Sigma \end{array}\begin{array}{c} \begin{array}{ccccc} x_1 & x_2 & x_6 & x_4 & b \end{array}\\ \left(\begin{array}{cccc|c} \frac{1}{3} & \frac{5}{3} & 0 & \frac{2}{3} & 6\\ \frac{2}{3} & \frac{1}{3} & 1 & \frac{1}{3} & 6\\ -\frac{1}{3} & -\frac{5}{3} & 0 & -\frac{2}{3} & -6 \end{array}\right)\end{array} \begin{array}{l} \\ \cdot\frac{3}{5}\ :5\\ \\ \\ \end{array} =$$

$$\begin{array}{c} \\ x_2\\ x_3\\ -\Sigma \end{array}\begin{array}{c} \begin{array}{ccccc} x_1 & x_5 & x_6 & x_4 & b \end{array}\\ \left(\begin{array}{cccc|c} \frac{1}{5} & 1 & 0 & \frac{2}{5} & \frac{18}{5}\\ \frac{3}{5} & 0 & 1 & \frac{1}{5} & \frac{24}{5}\\ 0 & 0 & 0 & 0 & 0 \end{array}\right)\end{array} = \begin{array}{c} \\ x_2\\ x_3\\ -\Sigma \end{array}\begin{array}{c} \begin{array}{ccc} x_1 & x_4 & b \end{array}\\ \left(\begin{array}{cc|c} \frac{1}{5} & \frac{2}{5} & \frac{18}{5}\\ \frac{3}{5} & \frac{1}{5} & \frac{24}{5}\\ 0 & 0 & 0 \end{array}\right)\end{array}$$

$x_2+\frac{1}{5}x_1+\frac{2}{5}x_4=\frac{18}{5}, x_2=\frac{18}{5}-\frac{1}{5}x_1-\frac{2}{5}x_4$

$x_3+\frac{2}{5}x_1+\frac{1}{5}x_4=\frac{24}{5}; x_3=\frac{24}{5}-\frac{3}{5}x_1-\frac{1}{5}x_4$

$$=-4x_1-6x_2-5x_3-3x_4=-4x_1-6\left(\frac{18}{5}-\frac{1}{5}x_1-\frac{2}{5}x_4\right)-5\left(\frac{24}{5}-\frac{3}{5}x_1-\frac{1}{5}x_4\right)-3x_4=$$

$$=-4x_1-\frac{108}{5}+\frac{6}{5}x_1+\frac{12}{5}x_4-24+3x_1+x_4-3x_4=\frac{1}{5}x_1+\frac{17}{5}x_4-\frac{228}{5}$$

$$\begin{array}{c} \\ x_2\\ x_3\\ F \end{array}\begin{array}{c} \begin{array}{ccc} x_1 & x_4 & b \end{array}\\ \left(\begin{array}{cc|c} \frac{1}{5} & \frac{2}{5} & \frac{18}{5}\\ \frac{3}{5} & \frac{1}{5} & \frac{24}{5}\\ \frac{1}{5} & \frac{17}{5} & \frac{228}{5} \end{array}\right)\end{array}$$

$x_2=\frac{18}{5}; x_3=\frac{24}{5}; x_1=0; x_4=0$

$F=\frac{228}{5}$

$$\begin{cases} 15x_1+12x_2+10x_3 \to max \\ x_1+x_2+x_3 \leq 100 \\ x_1 \geq 20 \\ x_2+x_3 \leq 60 \\ x_1 \ldots x_3 \geq 0 \end{cases}$$

$$\begin{cases} -15x_1-12x_2-10x_3 \to min \\ x_1+x_2+x_3+x_4 = 100 \\ x_1 - x_5 = 20 \\ x_2+x_3+x_6 = 60 \\ x_1 \ldots x_6 \geq 0 \end{cases}$$

$$\begin{cases} x_7+x_8+x_9 \to min \\ x_1+x_2+x_3+x_4+x_7 = 100 \\ x_1 - x_5 + x_8 = 20 \\ x_2+x_3+x_6+x_9 = 60 \\ x_1 \ldots x_9 \geq 0 \end{cases}$$

	x_1	x_2	x_3	x_4	x_5	x_6	b
x_7	1	1	1	1	0	0	100
x_8	1	0	0	0	-1	0	20
x_9	0	1	1	0	0	1	60
$-\Sigma$	-2	-2	-2	-1	1	-1	-180

=

	x_8	x_2	x_3	x_4	x_5	x_6	b
x_7	0	1	1	1	1	0	80
x_1	1	0	0	0	-1	0	20
x_9	0	1	1	0	0	1	60
$-\Sigma$	0	-2	-2	-1	-1	-1	-140

=

	x_8	x_9	x_3	x_4	x_5	x_6	b
x_7	0	0	0	1	1	-1	20
x_1	1	0	0	0	-1	0	20
x_2	0	1	1	0	0	1	60
$-\Sigma$	0	0	0	-1	-1	1	-20

=

	x_8	x_9	x_3	x_7	x_5	x_6	b
x_4	0	0	0	1	1	-1	20
x_1	1	0	0	0	-1	0	20
x_2	0	1	1	0	0	1	60
$-\Sigma$	0	0	0	0	0	0	0

=

	x_3	x_5	x_6	b
x_4	0	1	-1	20
x_1	0	-1	0	20
x_2	1	0	1	60
$-\Sigma$	0	0	0	0

$x_1 - x_5 = 20$; $x_1 = 20 + x_5$

$x_2 + x_3 + x_6 = 60$; $x_2 = 60 - x_3 - x_6$

$$-15x_1-12x_2-10x_3 = -15(20+x_5)-12(60-x_3-x_6)-10x_3 = -300-15x_5-720+12x_3+12x_6-10x_3 =$$
$$= 2x_3 - 15x_5 + 12x_6 - 1020$$

	x_3	x_5	x_6	b
x_4	0	1	-1	20
x_1	0	-1	0	20
x_2	1	0	1	60
F	2	-15	12	1020

=

	x_3	x_4	x_6	b
x_5	0	1	-1	20
x_1	0	0	-1	40
x_2	1	0	1	60
F	2	0	-3	1320

=

	x_3	x_4	x_2	b
x_5	1	1	0	80
x_1	1	0	0	100
x_6	1	0	1	60
F	5	0	0	1500

$x_5 = 80$; $x_1 = 100$; $x_6 = 60$

$x_2 = 0$; $x_3 = 0$; $x_4 = 0$ $\quad F = 1500$

$$\begin{cases} 3x_1+5x_2+2x_3 \to \min \\ 2x_1+x_2+x_3 \leq 10 \\ x_1 \geq 2 \\ x_3 \geq 1 \\ x_1 \dots x_3 \geq 0 \end{cases} \qquad \begin{cases} 3x_1+5x_2+2x_3 \to \min \\ 2x_1+x_2+x_3+x_4 = 10 \\ x_1 - x_5 = 2 \\ x_3 - x_6 = 1 \\ x_1 \dots x_6 \geq 0 \end{cases} \qquad \begin{cases} x_7+x_8+x_9 \to \min \\ 2x_1+x_2+x_3+x_4+x_7 = 10 \\ x_1 - x_5 + x_8 = 2 \\ x_3 - x_6 + x_9 = 1 \\ x_1 \dots x_9 \geq 0 \end{cases}$$

$$\begin{array}{c|cccccc|c} & x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & b \\ \hline x_7 & 2 & 1 & 1 & 1 & 0 & 0 & 10 \\ x_8 & 1 & 0 & 0 & 0 & -1 & 0 & 2 \\ x_9 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \\ -\Sigma & -3 & -1 & -2 & -1 & 1 & 1 & -13 \end{array} = \begin{array}{c|cccccc|c} & x_8 & x_2 & x_3 & x_4 & x_5 & x_6 & b \\ \hline x_7 & 0 & 1 & 1 & 1 & 2 & 0 & 6 \\ x_1 & 1 & 0 & 0 & 0 & -1 & 0 & 2 \\ x_9 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \\ -\Sigma & 0 & -1 & -2 & -1 & -2 & 1 & -7 \end{array} =$$

$$\begin{array}{c|cccccc|c} & x_8 & x_2 & x_9 & x_4 & x_5 & x_6 & b \\ \hline x_7 & 0 & 1 & 0 & 1 & 2 & 1 & 5 \\ x_1 & 1 & 0 & 0 & 0 & -1 & 0 & 2 \\ x_3 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \\ -\Sigma & 0 & -1 & 0 & -1 & -2 & -1 & -5 \end{array} \qquad \begin{array}{c|cccccc|c} & x_8 & x_2 & x_9 & x_4 & x_7 & x_6 & b \\ \hline x_5 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 1 & \frac{1}{2} & \frac{5}{2} \\ x_1 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & \frac{1}{2} & \frac{9}{2} \\ x_3 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \\ -\Sigma & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$